Beyond the Syllabus

PSPACE, Games, and Fixed-Parameter Tractability: From Quantum Provers to the Frontiers of FPT

Week 13 Supplement – TOAE209/TOAH209: Design and Analysis of Algorithms

Foreign Trade University

Outline

Why look beyond the syllabus?

- This week you saw that two-player games are naturally **PSPACE-complete** (the $\exists\forall\exists\forall\cdots$ structure of QSAT mirrors minimax), and that even though **Vertex Cover** is NP-hard, a *bounded search tree* makes it **fixed-parameter tractable**: $O(2^k n)$.
- Three questions a curious student should ask next:
  1. A single all-powerful prover convinces a verifier of a PSPACE claim (that is what a game *is*). What happens if you add a **second prover** who cannot talk to the first – does that make the verifier *more* powerful, or does it not matter?
  2. Vertex Cover's branching trick rescued an NP-hard problem via FPT. Does the *same* kind of trick always rescue a hard **game**, parameterized by how many moves are left?
  3. Is FPT theory a finished 1990s toolbox, or are genuinely new FPT algorithms still being discovered today?

How this supplement is different from a survey

Every claim below is sourced to a specific, verifiable paper (arXiv id / DOI / venue), not a vague "recent research shows."

Recap: a single prover is exactly PSPACE

- An **interactive proof**: a verifier (polynomial time, can flip coins) exchanges messages with a computationally unbounded **prover** who is trying to convince it that a claim is true. This is precisely the shape of a two-player game with the verifier as one player probing the prover's "winning strategy."
- **Theorem (Shamir, 1992): IP = PSPACE**. A single prover, interacting with a randomized polynomial-time verifier, can prove *exactly* the claims decidable in polynomial space – no more, no less.
- This is exactly this week's QSAT/minimax intuition, made formal: the prover plays the role of "the side trying to prove the formula true," and the verifier's questions play the role of the alternating quantifiers.

Add a second, non-communicating prover: jump to NEXP

- Now give the verifier **two** provers, kept in separate rooms so they cannot coordinate their answers once questioning starts (they may agree on a strategy beforehand, but cannot communicate *during* the protocol).
- **Theorem (Babai, Fortnow & Lund, 1991): MIP $=$ NEXP**. Cross-examining two provers who cannot coordinate turns out to be *strictly more powerful* than one prover alone – it captures nondeterministic *exponential*-time claims, believed to be far beyond PSPACE.
- Intuition: the verifier can ask the two provers *correlated* questions and catch any inconsistency between their answers, effectively forcing them to commit to a single, globally consistent exponentially-large object – something one prover alone can bluff around, but two isolated provers cannot.

Now let the provers share quantum entanglement: $\mathrm{MIP}^* = \mathrm{RE}$

- Keep the two provers unable to communicate, but let them share **quantum entanglement** prepared before the protocol starts (entanglement cannot transmit information, so this seems like it should not help).
- **Theorem (Ji, Natarajan, Vidick, Wright & Yuen, 2020): $\mathbf{MIP}^* = \mathbf{RE}$**. Astonishingly, entangled provers push the class all the way up to **RE** – exactly the languages decidable by the *Halting Problem*. There exist quantum multi-prover games that encode *"does this Turing machine halt?"* in whether the verifier accepts.
- arXiv:2001.04383; published as *MIP* = RE*, **Communications of the ACM**, 2021. As a corollary, the paper **refutes** the 1976 Connes Embedding Problem (an open question in operator-algebra theory) and resolves Tsirelson's problem in quantum information theory – a rare case of a complexity proof settling open problems in pure mathematics and physics.

The hierarchy, at a glance

Protocol	Power	What changed
1 prover	**PSPACE**	this week's games (Shamir, 1992)
2 provers, classical, isolated	**NEXP**	cross-examine for consistency (BFL, 1991)
2 provers, quantum-entangled	**RE**	entanglement, paradoxically, *helps* (JNVWY, 2020)

The lesson

"How many provers" and "can they share entanglement" are not footnotes – they are knobs that move the power of a game-like protocol from *this week's* PSPACE all the way to *undecidable*. The game-theoretic flavor you learned this week is the same flavor driving one of the deepest results in 21st-century theoretical computer science.

Recap: the bounded search tree that rescued Vertex Cover

- This week's FPT algorithm for Vertex Cover: pick any uncovered edge (u, v), branch on "$u \in C$" or "$v \in C$", decrement the budget k. Depth $\leq k$, branching factor 2 $\Rightarrow O(2^k n)$.
- The trick generalizes to many NP-hard problems: *find a small, local branching rule whose depth depends only on* k, and the exponential blow-up is confined to $f(k)$, however you slice the input size n.
- Natural question: if a **game** is parameterized by "how many moves remain," does an analogous branching trick always give an FPT algorithm?

No: Short Generalized Hex is provably *not* FPT

- **Short Generalized Hex:** given a Hex position on an arbitrary graph and an integer k, does the player to move have a winning strategy using at most k more moves?
- **Theorem (Bonnet, Gaspers, Lambilliotte, Rümmele & Saffidine, ICALP 2017):** Short Generalized Hex is **W[1]-complete** parameterized by k – resolving an open problem posed by Downey & Fellows in 1999.
- **W[1]** is the parameterized-complexity analogue of NP: a **W[1]**-hardness proof is a reduction-based lower bound showing that *no* $f(k) \cdot \mathrm{poly}(n)$ algorithm can exist, for *any* function f, unless the widely-believed conjecture **FPT** $\neq$ **W[1]** is false (the parameterized analogue of **P** $\neq$ **NP**).

Why Vertex Cover's trick does not transfer

- In Vertex Cover, branching on an edge's two endpoints *strictly shrinks* the remaining instance in a way that depends *only* on k: after k branches, either you have covered everything or you have failed – the branching factor and depth never depend on n.
- In Short Generalized Hex, the k "moves left" interact with the *entire board* at every step: a move's consequences can ripple through connections that scale with n, so no local branching rule can be found whose cost depends on k alone – the **W**[**1**]-hardness reduction makes this rigorous by encoding an arbitrary **W**[**1**]-hard instance's n-dependence *into* the k moves themselves.
- **Takeaway:** "parameterize by a small number" is necessary but not sufficient for FPT – you still need a genuine structural reason the exponential cost can be confined to that parameter, and sometimes (as here) it provably cannot.

FPT is not a finished 1990s toolbox

- Bounded search trees, kernelization, and color-coding are decades old, but **new FPT algorithms for new problems** are still appearing at top venues every year – including problems motivated by very recent applications.
- **Hedge Cut:** a generalization of graph cut where edges are grouped into *hedges*; cutting a hedge means removing *all* of its edges at once, at a single unit cost – modeling, e.g., a fiber-optic conduit whose failure severs many logical network links simultaneously.
- Prior work had shown Hedge Cut has **no polynomial-time algorithm** under the Exponential Time Hypothesis, and separately that it is **quasi-polynomial-time** solvable – leaving its fixed-parameter status open.

Hedge Cut is FPT (Fomin, Golovach, Korhonen, Lokshtanov & Saurabh, SODA 2025)

- **Theorem:** Hedge Cut is fixed-parameter tractable, parameterized by the solution size k (the number of hedges cut). arXiv:2410.17641; appeared at SODA 2025.
- Unlike Vertex Cover's single-edge branching rule, a hedge can bundle arbitrarily many graph edges together, so the "pick an edge, branch on its two endpoints" idea does not directly apply – the authors needed genuinely new combinatorial machinery (built on submodular-function and representative-set techniques from the FPT literature) to confine the exponential blow-up to k alone.
- This is the same overall *shape* of result as this week's Vertex Cover – "NP-hard in general, FPT in the natural parameter" – obtained for a problem that is structurally harder, using research published in the year before this course ran.

Summary

1. **Games scale far past PSPACE:** one prover gives PSPACE (this week); two isolated classical provers give NEXP (Babai–Fortnow–Lund, 1991); two quantum-entangled provers give **RE** – the power of the Halting Problem (Ji–Natarajan–Vidick–Wright–Yuen, 2020), incidentally refuting a 1976 open problem in pure mathematics.
2. **FPT's branching trick is not automatic:** it rescues Vertex Cover, but Short Generalized Hex is provably **W**[**1**]-complete (Bonnet et al., ICALP 2017) – some parameterizations of some games can never be made FPT, absent a collapse of **FPT** and **W**[**1**].
3. **FPT research is active right now:** Hedge Cut's fixed-parameter tractability (Fomin, Golovach, Korhonen, Lokshtanov & Saurabh, SODA 2025) needed new ideas beyond this week's bounded-search-tree template, published the year before this course ran.

Looking ahead

Next week (Kleinberg & Tardos, Chapters 11–12) turns from *exact* algorithms for hard problems to **approximation** and **local search** – trading the exponential cost of exactness for a provable closeness to optimal in polynomial time.